A Three levels of an information-processing system (Marr, 1982)

The same table describes a brain or a machine. Dashed arrows: agreement on one row does not fill the rows below.

	Brain *biological substrate*	**Machine** *artificial substrate*
1 Computational theory *this paper: function* What is computed, and why that mapping is appropriate	Prediction that keeps a body viable. Agency and value are tied to need.	Next-token prediction on a training corpus. No survival-tied objective.
loosely coupled		
2 Representation and algorithm *this paper: dynamics* Which internal variables, and how they are transformed	Multistable dynamics; attractors. Error units; hierarchical inference.	Gradient flow on a loss landscape. Tokens, embeddings, attention.
loosely coupled		
3 Hardware implementation *this paper: architecture* How representation and algorithm are realized physically	Neurons, synapses, spikes. Energy, and a body.	Weights, matrix multiplies, silicon. No metabolism, no body.

Not a Marr level

Theme (ii) starts from the bottom row and leaves it.
Viability, phenomenal experience, and values grounded in need are not properties of a hardware description.